

ABSTRACT & FORMATION DRILLING WINDOWS

This paper integrates wireline log signatures (GR, RHOB, DTC) and historical offset well data across 45 wells to construct a predictive geomechanical model.

Formation Name	Depth Interval	Pore Pressure (EMW)	Fracture Gradient	Primary Drilling Hazard
Tipam Sandstone	0 - 800 m	1.02 - 1.08 g/cc	1.65 g/cc	Washout, high permeability
Girujan Clay	800 - 1,500 m	1.10 - 1.15 g/cc	1.72 g/cc	Clay swelling, tight hole
Barail Group	1,800 - 2,500 m	1.18 - 1.25 g/cc	1.80 g/cc	■ Reactive Shale / Stuck Pipe
Kopili Formation	2,500 - 3,200 m	1.28 - 1.38 g/cc	1.88 g/cc	■ Overpressure Kick / Influx
Sylhet Limestone	3,000 - 3,500 m	1.15 - 1.22 g/cc	1.52 g/cc	■ Fractured Lost Circulation